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Concepts

Training and Evaluation

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Data Preparation

Module 9 Introduction

- Introduction to Module 9's Study
Video • 2 min
- Why Data Preparation Is Not Optional in
Machine Learning
Reading • 5 min

Module 9-1 Basic Data Cleaning Review

- ✓ **Basic Data Cleaning Review**
Video • 38 sec
- ✓ **Materials on Basic Data Cleaning**
Reading • 10 min
- ✓ **Cleaning Data Is a Decision, Not a Checklist**
Reading • 5 min

Module 9-2 Distributions, Outliers and Their Removal

- Distributions, Outliers and Their Removal:
Gaussian Distribution, Skewness and Outliers
Video • 6 min
- Distributions, Outliers and Their Removal: The
Z-Score and IQR Method for Outliers Removal
Video • 5 min
- Materials on Distributions, Outliers and Their
Removal
Reading • 10 min
- Outliers, Assumptions, and When "Cleaning"
Becomes Damage
Reading • 5 min

Module 9-3 Data Transform

- **Data Transform: Normalization, Standardization, Power Transform**
Video • 7 min
- **Materials on Data Transform**
Reading • 10 min
- **When Scaling Helps — and When It Quietly Breaks Your Model**
Reading • 5 min

Evaluation

- Construct Training and Test Sets for Model Evaluation: Methods, Implementation, and Stratified Sampling
Video • 5 min
- Construct Training and Test Sets for Model Evaluation: Cross-Validation
Video • 6 min
- Materials on Training/Test Sets Generation
Reading • 10 min
- Evaluation Starts with the Split: When Performance Numbers Lie
Reading • 5 min

Module 9-5 Data Preparation Overview

- **Data Preparation Overview**
Video • 3 min
- **Materials on Data Preparation**
Reading • 10 min
- **From Raw Data to Reliable Learning: What Actually Matters?**
Reading • 5 min
- **Data Preparation and Evaluation Coach**
Dialogue • 5 min

Module 9 Quiz

- Module 9 Quiz
Practice Assignment • 30 min

Module 9 Assignments

- Module 9 Assignment 1
Graded App Item • 30 min
- Module 9 Assignment 2
Graded App Item • 30 min
- Module 9 Assignment 3
Graded App Item • 30 min
- XLS file for Module 9 Assignment
Reading • 1 min
- Module 9 Recap
Reading • 5 min

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Module 10 Introduction

- Introduction to Module 10's Study Video • 1 min
- Why Support Vector Machines? Reading • 5 min

Module 10-1 Support Vector Machine Fundamentals

- Support Vector Machine Fundamentals:
 Contents

Cleaning Data Is a Decision, Not a Checklist

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1. Reflecting on Identifying Data Problems

- Reflect on an example of incorrect data, missing values, or duplicate records that you have encountered—or could realistically expect to encounter—in a scientific, engineering, or research dataset.
- Consider how leaving this issue unresolved could affect model training, evaluation, or the reliability of predictions.

2. Reflecting on Cleaning Choices and Their Consequences

- When identifying missing or incorrect values in a dataset, reflect on the possible strategies available besides simply deleting the affected data.
- Consider situations where removing too much data during cleaning could negatively impact the dataset, reduce important variability, or distort the learning process.

3. Reflecting on Data Cleaning vs. Data Distortion

- Reflect on whether data cleaning itself can unintentionally alter the original data distribution in harmful ways.
- Consider how certain cleaning decisions could introduce bias, remove meaningful rare events, or create misleading patterns in the dataset.

4. Reflecting on the Limits of Basic Data Cleaning

- Reflect on why basic cleaning steps such as handling missing values, correcting errors, and removing duplicates are necessary but still insufficient for preparing data for machine learning.
- Consider what limitations of basic cleaning may require additional techniques such as distribution analysis, outlier handling, scaling, normalization, or data transformation.

✓ Completed

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